

Cryogenic Device Consequences: Persistent Trapping and Slower Recovery

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This note connects the cryogenic defect-kinetics discussion to a concrete device-level consequence. The main idea is that radiation-created traps may not relax quickly at low temperature. Instead, trapped charge and damage signatures can persist over long operational timescales.

Physical picture

At room temperature, some trapped carriers can be thermally re-emitted into the bands, and some defect populations can partially recover through annealing. As temperature is lowered, both processes slow down strongly. The result is that the device may retain a memory of prior irradiation through persistent trapped charge or slow defect recovery.

Simple recovery model

A simple phenomenological model for the trapped-charge recovery is

$$Q_{\text{trap}}(t, T) = Q_0 \exp\left(-\frac{t}{\tau_{\text{rec}}(T)}\right),$$

where Q_0 is the initial trapped-charge population and $\tau_{\text{rec}}(T)$ is a temperature-dependent recovery time constant.

A first-order Arrhenius model for the recovery time constant is

$$\tau_{\text{rec}}(T) = \tau_0 \exp\left(\frac{E_a}{k_B T}\right).$$

As T decreases, the recovery time can become extremely large.

Interpretation

This model is intentionally simple, but it captures the main cryogenic trend:

- at higher temperature, recovery can occur on experimentally accessible timescales,
- at lower temperature, recovery slows dramatically,
- at deep cryogenic temperature, the device may effectively preserve trapped charge and radiation history during operation.

Engineering implication

A cryogenic silicon device is not simply a room-temperature device with lower thermal noise. Radiation-induced trap populations may persist much longer, which can affect offset drifts, transient recovery, charge collection history, and reproducibility after irradiation.

Positioning

This section should be presented as a first-order kinetics extension rather than a full microscopic model. It is useful because it converts the cryogenic defect story into one clear device consequence that is easy to communicate.